SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5111

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Kala Gowri Shankar(192325109) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message.

b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

"GSRH RH Z HVXIVG"

This is most likely encrypted using the **Atbash Cipher**, where:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- E ↔ V
- F ↔ U
- G ↔ T
- H ↔ S
- ...and so on.

a. Original Message

Using the Atbash Cipher:

Cipher Text	**Plain Text**
G	T
S	H
R	I
H	S
R	I
H	S
Z	A
H	S
V	E
X	C
I	R
V	E
G	T

Original Message:

"THIS IS A SECRET"

b. Justification

The message is likely encrypted using the **Atbash Cipher** because:

1. **Simple substitution cipher**
    - Each letter is replaced with its opposite letter in the alphabet.
    - Example:
        - A → Z
        - B → Y
        - C → X
        - T → G
        - H → S
2. **Repeating pattern**
    - The word **"RH"** appears twice.
    - After decoding, it becomes **"IS"**, a common English word.
3. **Letter structure**
    - The first four letters **"GSRH"** decode to **"THIS"**, which is a meaningful English word.
    - The last word **"HVXIVG"** decodes to **"SECRET"**, forming the sentence: **"THIS IS A SECRET"**
4. **Meaningful sentence**
    - After decoding, the message becomes a clear English sentence, confirming that the Atbash Cipher was used.

Final Answer (Exam Format)

a. Original Message

THIS IS A SECRET

b. Justification

- The message is encrypted using the **Atbash Cipher**, a simple substitution cipher.
- In the Atbash Cipher, each letter is replaced by its opposite letter in the alphabet (A↔Z, B↔Y, C↔X, etc.).
- The repeated word **"RH"** becomes **"IS"**, showing a common English word.
- The first word **"GSRH"** becomes **"THIS"**, and the last word **"HVXIVG"** becomes **"SECRET"**.
- Therefore, the decoded message is **"THIS IS A SECRET."**

2.Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a. Analyze how investigators could detect the presence of hidden data within the image.

b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

a. How investigators can detect hidden data in the image

Investigators can use the following methods:

1. **Use Steganography Detection Software**
    - They use tools like StegExpose or StegDetect to check if hidden data is inside the image.
2. **Analyze Image Properties**
    - They check the image size, file format, and metadata.
    - If the file size is **unusually large, it may contain hidden information.**
3. **Compare Pixel Values**
    - They examine the image pixels.
    - Hidden data often changes small pixel values that are not visible to the human eye.
4. **Check Metadata**
    - They inspect the image metadata (date, author, comments).
    - Hidden information may be stored there.

b. Two techniques to improve the security of hidden communication

1. Encrypt the Data Before Hiding

- First, convert the secret message into encrypted text using an encryption algorithm (such as AES).
- Then hide the encrypted data inside the image.
- Even if someone finds the hidden data, they cannot read it without the encryption key.

2. Use Advanced Steganography Techniques

- Hide the data in different parts of the image using random locations or stronger algorithms instead of placing it in predictable positions.
- This makes the hidden information much harder to detect or extract.

Final Answer

a. Detection of Hidden Data

- Use steganography detection tools.
- Analyze image size, format, and metadata.
- Compare pixel values for unusual changes.
- Check metadata for hidden information.

b. Techniques to Improve Security

1. Encrypt the secret data before hiding it in the image.
2. Use advanced steganography methods that hide data in random or less predictable locations.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25**	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies.	Good understanding with proper integration of most concepts.	Basic understanding with limited or partial integration.	Poor understanding; unable to integrate concepts.
Experimental Design	**30**	Well-planned, systematic implementation with an optimal solution.	Correct implementation with minor issues.	Basic implementation with errors or inefficiencies.	Incorrect or incomplete implementation.
Use of Tools	**20**	Effective and advanced use of tools/technologies with proper configuration.	Appropriate use of tools with correct execution.	Limited or inconsistent use of tools.	Incorrect or minimal use of tools.
Analysis of Results	**15**	Insightful analysis with accurate interpretation and validation of results.	Good analysis with reasonable interpretation.	Basic analysis with limited insights.	Poor or incorrect analysis of results.
Documentation	**10**	Clear, well-structured documentation with diagrams, code, and results.	Organized documentation with necessary details.	Basic documentation with missing details.	Poor or incomplete documentation.